Lec-5
[illegible]

# Data Types & Type Casting

9.4

① Number-3 → Mem.

① Number System Data Types and Type Casting

→ Primitive
→ Non Primitive

int x=5 byte b=5

byte
short
int
long
float
double
char
boolean

Range

| | |
|---|---|
| byte | 8bit → 1byte |
| short | 16bit → 2byte |
| int | 32bit → 4byte |
| long | 64bit → 8byte |
| float | 32bit → 4byte |
| double | 64bit → 8byte |
| char | 16bit → 2byte |
| boolean | 1bit |

[ - - - - - - - ]
8bit → byte

( - - - - - - - ) 2^8

Number System

10 decimal → 9216
2 Binary → 2^14
8 Octal → 2^12
16 Hexa → 2^8

9x4 · 4x5
3x2

| | | |
|---|---|---|
| 2 | 33 | 1 x1=1 |
| 2 | 16 | 0 x2 |
| 2 | 8 | 1 x4 |
| 2 | 4 | 0 x8 |
| 2 | 2 | 1 x16 |
| 2 | 1 | 1 x32 |
| | 0 | |

while(n>0){
sum = n%2
sum = sum*powerof2
n = n/2
}

sum = n%10
sum = sum*powerof2
n = n/10

| | | |
|---|---|---|
| 10 | 101011 | 1 x1 |
| 10 | 10101 | 0 x2 |
| 10 | 10111 | 1 x4 |
| 10 | 10101 | 0 x8 |
| 10 | 1010 | 1 x16 |
| 10 | 101 | 0 x32 |
| 10 | 10 | 1 x64 |
| 10 | 1 | 0 x128 |
| | 0 | 1 x256 |

(101011)
0 1 2 3 4 5

1x2^0 + 1x2^1 + 1x2^2 + 1x2^4 + 1x2^5

10 (1010)

21337

5 x 10 → 50 → 0

| | | |
|---|---|---|
| 10 | 2133 | 7 |
| 10 | 213 | 3 |
| 10 | 21 | 1 |
| | 2 | |

while(n>0){
sum = n%10
sum = sum*powerof2
mul = mul*10
n = n/10
}

Range

| | |
|---|---|
| byte | 8bit → 1byte |
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| boolean | 1bit |

101

byte b=5

0 0 0 0 0 1 0 1

101 → 00000101

2^7

[ 0 0 0 0 0 0 0 0 ]

2^8 = 256

[-x]

2's complement

[ 0 | 1 1 1 1 1 1 1 ]

[-128, 127] → [-127]

[ 0 0 0 0 0 0 0 0 ] → 0

2's complement

0 1 0 1 0 1 0 1

1 0 1 0 1 0 1 0

0 1 1 1 1 1 1 1

2's complement

0 1 0 1 0 1 1 1

1 0 1 0 1 0 0 0

0 0 0 0 0 0 0 1

x = 107

1 0 0 1 0 1 0 1 = -101

[-128 127]

byte

0 1 0 1 0 1 1 1

1 0 0 0 0 0 0 0

0 1 1 1 1 1 1 1

0000 0000

0 1 1 1 1 1 1 1

0000 0000

byte = -128 + 127 = -1

byte → 8bit → 1byte → $-2^{8-1}$ to $2^{8-1}-1$ → $-2^7$ to $2^7-1$
short → 16 bit → 2byte → $-2^{15}$ to $2^{15}-1$
int → 32 bit → 4byte → $-2^{31}$ to $2^{31}-1$
long → 64 bit → 8byte → $-2^{63}$ to $2^{63}-1$

x

1 0 1 0 1 0 0 0

1 0 1 0 1 0 0 0

0 0 1 0 1 0 1 1

1 0 1 0 1 0 0

0 1 0 1 0 0 1

1 1 1

0 1 0 1 0 0 0

0 0 0 0 1 0 1

byte b=5

0 0 0 0 1 0 1

1 1 1 1 0 1 1

1 1 1 1 0 1 1

int x=5;

int x=5;

0 0 0 0 1 0 1

0 0 0 0 1 0 1

0 0 0 0 1 0 1

1 1 1 1 0 1 1

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
// byte b = (byte) (420);// 164
    byte b = (byte) (300);//
    short s = 5;
    int n = 5;
    long l = 5;
    System.out.println(n);
    int[]
    int[]
    long l1= 67687983491;
}
```

int

long

5kB

int[]

int[]

int[]